

# 重庆邮电大学线性代数习题册详解

## 第1章 行列式

### 习题 1.1

1. 利用对角线法则计算下列三阶行列式:

$$(1) \begin{vmatrix} 2 & 0 & 1 \\ 1 & -4 & -1 \\ -1 & 8 & 3 \end{vmatrix};$$

$$(2) \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix}.$$

$$\text{解 (1)} \begin{vmatrix} 2 & 0 & 1 \\ 1 & -4 & -1 \\ -1 & 8 & 3 \end{vmatrix} = 2 \times (-4) \times 3 + 0 \times (-1) \times (-1) + 1 \times 1 \times 8$$

$$-1 \times (-4) \times (-1) - 0 \times 1 \times 3 - 2 \times (-1) \times 8 = -4.$$

$$\begin{aligned} (2) \begin{vmatrix} x & y & x+y \\ y & x+y & x \\ x+y & x & y \end{vmatrix} &= xy(x+y) + xy(x+y) + xy(x+y) - (x+y)^3 - x^3 - y^3 \\ &= -2x^3 - 2y^3. \end{aligned}$$

2. 用行列式解线性方程组

$$\begin{cases} x_1 + x_2 - 2x_3 = -3, \\ x_1 - 2x_2 + 2x_3 = 2, \\ 2x_1 - 5x_2 + 4x_3 = 4. \end{cases}$$

**解** 方程组的系数行列式

$$D = \begin{vmatrix} 1 & 1 & -2 \\ 1 & -2 & 2 \\ 2 & -5 & 4 \end{vmatrix} = 4, \quad D_1 = \begin{vmatrix} -3 & 1 & -2 \\ 2 & -2 & 2 \\ 4 & -5 & 4 \end{vmatrix} = -2,$$

$$D_2 = \begin{vmatrix} 1 & -3 & -2 \\ 1 & 2 & 2 \\ 2 & 4 & 4 \end{vmatrix} = 0, \quad D_3 = \begin{vmatrix} 1 & 1 & -3 \\ 1 & -2 & 2 \\ 2 & -5 & 4 \end{vmatrix} = 5.$$

得方程组的解为  $x_1 = -\frac{1}{2}, x_2 = 0, x_3 = \frac{5}{4}$ .

### 习题 1.2

1. 计算下列排列的逆序数, 并判断排列的奇偶性:

$$(1) 3412; \quad (2) 24315876;$$

解(1)  $t(3412) = 4$ , 偶排列; (2)  $t(24315876) = 7$ , 奇排列

2. 计算下列排列的逆序数.

(1)  $13 \cdots (2n-1)24 \cdots (2n)$ ; (2)  $13 \cdots (2n-1)(2n)(2n-2) \cdots 2$ .

解 (1)  $t(13 \cdots (2n-1)24 \cdots (2n)) = \frac{n(n-1)}{2}$ ;

(2)  $t(13 \cdots (2n-1)(2n)(2n-2) \cdots 2) = n(n-1)$ .

3. 选择  $i$  与  $j$  使下述 9 元排列

(1)  $1245i6j97$  为奇排列; (2)  $3972i51j4$  为偶排列.

解 (1)  $i = 8, j = 3$ ; (2)  $i = 8, j = 6$ .

### 习题 1.3

1. 设五阶行列式

$$D = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{15} \\ a_{21} & a_{22} & \cdots & a_{25} \\ \vdots & \vdots & & \vdots \\ a_{51} & a_{52} & \cdots & a_{55} \end{vmatrix},$$

则下列各式是否可能为  $D$  的项? 如果可能, 试确定它的正负号, 使其成为  $D$  的项:

(1)  $a_{31}a_{43}a_{21}a_{52}a_{55}$ ;

(2)  $a_{31}a_{23}a_{45}a_{12}a_{54}$ .

解 (1) 因  $a_{31}, a_{21}$  都取自第一列, 所以  $a_{31}a_{43}a_{21}a_{52}a_{55}$  不可能为  $D$  的项;

(2) 因  $a_{31}a_{23}a_{45}a_{12}a_{54}$  取自  $D$  的不同行不同列, 所以  $a_{31}a_{23}a_{45}a_{12}a_{54}$  为  $D$  的项, 又

$a_{31}a_{23}a_{45}a_{12}a_{54} = a_{12}a_{23}a_{31}a_{45}a_{54}$ , 而  $t(23154) = 3$ , 所以该项取负号.

2. 写出四阶行列式  $\begin{vmatrix} a_{11} & \cdots & a_{14} \\ \vdots & & \vdots \\ a_{41} & \cdots & a_{44} \end{vmatrix}$  中一切带有负号且含元素  $a_{23}$  的项.

解  $-a_{11}a_{23}a_{32}a_{44}$ ;  $-a_{12}a_{23}a_{34}a_{41}$ ,  $-a_{14}a_{23}a_{31}a_{42}$ .

3. 根据行列式定义计算  $n$  阶行列式:

$$\begin{vmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 2 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & n-1 \\ n & 0 & 0 & \cdots & 0 \end{vmatrix}.$$

解 由于行列式中的项形如  $(-1)^{t(p_1 p_2 \cdots p_n)} a_{1p_1} a_{2p_2} \cdots a_{np_n}$ , 当  $p_1 \neq 2$  时,  $a_{1p_1} = 0$ , 因此  $p_1$  只

能取 2, 同理  $p_2$  只能取 3,  $\cdots$ ,  $p_{n-1}$  只能取  $n$ ,  $p_n$  只能取 1, 因此, 不为零的项只有一项

$a_{12}a_{23}\cdots a_{n-1,n}a_{n1}$ , 即

$$D = (-1)^{(23\cdots n1)} a_{12}a_{23}\cdots a_{n-1,n}a_{n1} = (-1)^{n-1} 1 \times 2 \times \cdots \times (n-1) \times n = (-1)^{n-1} n!;$$

#### 习题 1.4

1. 利用行列式性质计算下列行列式.

$$(1) \begin{vmatrix} 3421 & 3521 \\ 2809 & 2909 \end{vmatrix}; \quad (2) \begin{vmatrix} -ab & ac & ae \\ bd & -cd & de \\ bf & cf & -ef \end{vmatrix}; \quad (3) \begin{vmatrix} 4 & 1 & 2 & 1 \\ 2 & 1 & 3 & -1 \\ 3 & 2 & 1 & 2 \\ 6 & 2 & 5 & 0 \end{vmatrix};$$

$$(4) \begin{vmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{vmatrix};$$

$$\text{解 } (1) \begin{vmatrix} 3421 & 3521 \\ 2809 & 2909 \end{vmatrix} \xrightarrow{c_2 - c_1} \begin{vmatrix} 3421 & 100 \\ 2809 & 100 \end{vmatrix} = 100 \begin{vmatrix} 3421 & 1 \\ 2809 & 1 \end{vmatrix} = 100(3421 - 2809) = 61200;$$

$$(2) \begin{vmatrix} -ab & ac & ae \\ bd & -cd & de \\ bf & cf & -ef \end{vmatrix} = abcdef \begin{vmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = abcdef \begin{vmatrix} -1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 2 & 0 \end{vmatrix} = 4abcdef;$$

$$(3) \begin{vmatrix} 4 & 1 & 2 & 1 \\ 2 & 1 & 3 & -1 \\ 3 & 2 & 1 & 2 \\ 6 & 2 & 5 & 0 \end{vmatrix} \xrightarrow{r_2 + r_1} \begin{vmatrix} 4 & 1 & 2 & 1 \\ 6 & 2 & 5 & 0 \\ 3 & 2 & 1 & 2 \\ 6 & 2 & 5 & 0 \end{vmatrix} = 0;$$

$$(4) \begin{vmatrix} 3 & 1 & 1 & 1 \\ 1 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 1 & 1 & 1 & 3 \end{vmatrix} = \begin{vmatrix} 6 & 1 & 1 & 1 \\ 6 & 3 & 1 & 1 \\ 6 & 1 & 3 & 1 \\ 6 & 1 & 1 & 3 \end{vmatrix} = \begin{vmatrix} 6 & 1 & 1 & 1 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{vmatrix} = 48;$$

$$2. \text{ 已知 } \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = m, \text{ 求 } \begin{vmatrix} 4a_{11} & 2a_{13} - 3a_{11} & -a_{12} \\ 4a_{21} & 2a_{23} - 3a_{21} & -a_{22} \\ 4a_{31} & 2a_{33} - 3a_{31} & -a_{32} \end{vmatrix} \text{ 的值.}$$

$$\text{解 } \begin{vmatrix} 4a_{11} & 2a_{13} - 3a_{11} & -a_{12} \\ 4a_{21} & 2a_{23} - 3a_{21} & -a_{22} \\ 4a_{31} & 2a_{33} - 3a_{31} & -a_{32} \end{vmatrix} = 4 \begin{vmatrix} a_{11} & 2a_{13} & -a_{12} \\ a_{21} & 2a_{23} & -a_{22} \\ a_{31} & 2a_{33} & -a_{32} \end{vmatrix} = 8 \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = 8m.$$

3. 证明:

$$(1) \begin{vmatrix} a^2 & ab & b^2 \\ 2a & a+b & 2b \\ 1 & 1 & 1 \end{vmatrix} = (a-b)^3; \quad (2) \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix} = 0;$$

$$(3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^4 & b^4 & c^4 & d^4 \end{vmatrix} = (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)(a+b+c+d)$$

$$\begin{aligned} \text{证 } (1) \begin{vmatrix} a^2 & ab & b^2 \\ 2a & a+b & 2b \\ 1 & 1 & 1 \end{vmatrix} &= \begin{vmatrix} a^2 & ab-a^2 & b^2-a^2 \\ 2a & b-a & 2b-2a \\ 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} ab-a^2 & b^2-a^2 \\ b-a & 2b-2a \end{vmatrix} \\ &= \begin{vmatrix} ab-a^2 & b^2-a^2 \\ b-a & 2b-2a \end{vmatrix} = (b-a)^2 \begin{vmatrix} a & b+a \\ 1 & 2 \end{vmatrix} = (b-a)^2(a-b) = (a-b)^3. \end{aligned}$$

$$\begin{aligned} (2) \begin{vmatrix} a^2 & (a+1)^2 & (a+2)^2 & (a+3)^2 \\ b^2 & (b+1)^2 & (b+2)^2 & (b+3)^2 \\ c^2 & (c+1)^2 & (c+2)^2 & (c+3)^2 \\ d^2 & (d+1)^2 & (d+2)^2 & (d+3)^2 \end{vmatrix} &= \begin{vmatrix} a^2 & 2a+1 & 4a+4 & 6a+9 \\ b^2 & 2b+1 & 4b+4 & 6b+9 \\ c^2 & 2c+1 & 4c+4 & 6c+9 \\ d^2 & 2d+1 & 4d+4 & 6d+9 \end{vmatrix} \\ &= \begin{vmatrix} a^2 & 2a+1 & 2 & 6 \\ b^2 & 2b+1 & 2 & 6 \\ c^2 & 2c+1 & 2 & 6 \\ d^2 & 2d+1 & 2 & 6 \end{vmatrix} = 0. \end{aligned}$$

$$\begin{aligned} (3) \begin{vmatrix} 1 & 1 & 1 & 1 \\ a & b & c & d \\ a^2 & b^2 & c^2 & d^2 \\ a^4 & b^4 & c^4 & d^4 \end{vmatrix} &= \begin{vmatrix} 1 & 1 & 1 & 1 \\ 0 & b-a & c-a & d-a \\ 0 & b(b-a) & c(c-a) & d(d-a) \\ 0 & b^2(b^2-a^2) & c^2(c^2-a^2) & d^2(d^2-a^2) \end{vmatrix} \\ &= \begin{vmatrix} b-a & c-a & d-a \\ b(b-a) & c(c-a) & d(d-a) \\ b^2(b^2-a^2) & c^2(c^2-a^2) & d^2(d^2-a^2) \end{vmatrix} \\ &= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 1 & 1 \\ b & c & d \\ b^2(b+a) & c^2(c+a) & d^2(d+a) \end{vmatrix} \\ &= (b-a)(c-a)(d-a) \begin{vmatrix} 1 & 0 & 0 \\ b & c-b & d-b \\ b^2(b+a) & (c-b)(c^2+b^2+bc+ac+ab) & (d-b)(d^2+b^2+bd+ab+ad) \end{vmatrix} \end{aligned}$$

$$\begin{aligned}
&= (b-a)(c-a)(d-a)(c-b)(d-b) \begin{vmatrix} 1 & 1 \\ c^2 + b^2 + bc + ac + ab & d^2 + b^2 + bd + ab + ad \end{vmatrix} \\
&= (b-a)(c-a)(d-a)(c-b)(d-b)[d^2 - c^2 + b(d-c) + a(d-c)] \\
&= (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)(a+b+c+d).
\end{aligned}$$

4. 计算下列  $n$  阶行列式的值.

$$(1) D_n = \begin{vmatrix} 1 & -1 & \cdots & -1 & -1 \\ 1 & 1 & \cdots & -1 & -1 \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & \cdots & 1 & -1 \\ 1 & 1 & \cdots & 1 & 1 \end{vmatrix};$$

$$(2) D_n = \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & & \vdots \\ a & a & \cdots & x \end{vmatrix},$$

$$(3) D_n = \begin{vmatrix} 1+a_1 & 1 & \cdots & 1 & 1 \\ 1 & 1+a_2 & \cdots & 1 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & \cdots & 1+a_{n-1} & 1 \\ 1 & 1 & \cdots & 1 & 1+a_n \end{vmatrix}, \text{ 其中 } a_1 a_2 \cdots a_n \neq 0;$$

$$(4) \begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_1 & a_2 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_2 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1-a_{n-1} & a_n \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{vmatrix}$$

$$\text{解 (1) } D_n = \begin{vmatrix} 1 & -1 & \cdots & -1 & -1 \\ 1 & 1 & \cdots & -1 & -1 \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & \cdots & 1 & -1 \\ 1 & 1 & \cdots & 1 & 1 \end{vmatrix} \xrightarrow{r_i - r_1 (i=2 \cdots n)} \begin{vmatrix} 1 & -1 & \cdots & -1 & -1 \\ 0 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 2 & \cdots & 2 & 0 \\ 0 & 2 & \cdots & 2 & 2 \end{vmatrix} = 2^{n-1}.$$

(2)

$$\begin{aligned}
 D_n &= \begin{vmatrix} x & a & \cdots & a \\ a & x & \cdots & a \\ \vdots & \vdots & \ddots & \vdots \\ a & a & \cdots & x \end{vmatrix} = \begin{vmatrix} x & a & a & \cdots & a \\ a-x & x-a & 0 & \cdots & 0 \\ a-x & 0 & x-a & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a-x & 0 & 0 & 0 & x-a \end{vmatrix} \\
 &= \begin{vmatrix} x+(n-1)a & a & a & \cdots & a \\ 0 & x-a & 0 & \cdots & 0 \\ 0 & 0 & x-a & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & x-a \end{vmatrix} \\
 &= [x+(n-1)a](x-a)^{n-1}.
 \end{aligned}$$

$$(3) \quad D_n = \begin{vmatrix} 1+a_1 & 1 & \cdots & 1 & 1 \\ 1 & 1+a_2 & \cdots & 1 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & \cdots & 1+a_{n-1} & 1 \\ 1 & 1 & \cdots & 1 & 1+a_n \end{vmatrix} \xrightarrow{r_i - r_1, i=2 \cdots n} \begin{vmatrix} 1+a_1 & 1 & \cdots & 1 & 1 \\ -a_1 & a_2 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ -a_1 & 0 & \cdots & a_{n-1} & 0 \\ -a_1 & 0 & \cdots & 0 & a_n \end{vmatrix}$$

$$= a_1 a_2 \cdots a_n \begin{vmatrix} 1+\frac{1}{a_1} & \frac{1}{a_2} & \cdots & \frac{1}{a_{n-1}} & \frac{1}{a_n} \\ -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ -1 & 0 & \cdots & 1 & 0 \\ -1 & 0 & \cdots & 0 & 1 \end{vmatrix} \xrightarrow{\frac{c_i}{a_i} (i=1 \cdots n)} \begin{vmatrix} 1+\sum_{i=1}^n \frac{1}{a_i} & \frac{1}{a_2} & \cdots & \frac{1}{a_{n-1}} & \frac{1}{a_n} \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & \cdots & 0 & 1 \end{vmatrix}$$

$$= (1 + \sum_{i=1}^n \frac{1}{a_i}) a_1 a_2 \cdots a_n.$$

(4) 依次进行第一行加到第二行，第二行加到第三行， $\cdots$ ，第  $n$  行加到第  $n+1$  行，得

$$\begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ -1 & 1-a_1 & a_2 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1-a_2 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1-a_{n-1} & a_n \\ 0 & 0 & 0 & 0 & \cdots & -1 & 1-a_n \end{vmatrix} = \begin{vmatrix} 1 & a_1 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 1 & a_2 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & a_3 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 1 & a_n \\ 0 & 0 & 0 & 0 & \cdots & 0 & 1 \end{vmatrix} = 1$$

### 习题 1.5

1. 计算下列行列式：

$$(1) \begin{vmatrix} x & y & 0 & 0 \\ 0 & x & y & 0 \\ 0 & 0 & x & y \\ y & 0 & 0 & x \end{vmatrix}; \quad (2) \begin{vmatrix} 2+x & 2 & 2 & 2 \\ 2 & 2-x & 2 & 2 \\ 2 & 2 & 2+y & 2 \\ 2 & 2 & 2 & 2-y \end{vmatrix};$$

解

$$(1) \begin{vmatrix} x & y & 0 & 0 \\ 0 & x & y & 0 \\ 0 & 0 & x & y \\ y & 0 & 0 & x \end{vmatrix} = x \begin{vmatrix} x & y & 0 \\ 0 & x & y \\ 0 & 0 & x \end{vmatrix} - y \begin{vmatrix} y & 0 & 0 \\ x & y & 0 \\ 0 & x & y \end{vmatrix} = x^4 - y^4.$$

$$(2) \begin{vmatrix} 2+x & 2 & 2 & 2 \\ 2 & 2-x & 2 & 2 \\ 2 & 2 & 2+y & 2 \\ 2 & 2 & 2 & 2-y \end{vmatrix} = \begin{vmatrix} x & x & 0 & 0 \\ 2 & 2-x & 2 & 2 \\ 0 & 0 & y & y \\ 2 & 2 & 2 & 2-y \end{vmatrix} = \begin{vmatrix} x & 0 & 0 & 0 \\ 2 & -x & 2 & 0 \\ 0 & 0 & y & 0 \\ 2 & 0 & 2 & -y \end{vmatrix}$$

$$= x \begin{vmatrix} -x & 2 & 0 \\ 0 & y & 0 \\ 0 & 2 & -y \end{vmatrix} = x^2 y^2.$$

2. 设

$$D = \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 2 & 3 & -4 \\ 2 & -1 & 1 & -1 \\ 1 & -5 & 3 & -3 \end{vmatrix},$$

计算  $A_{31} + A_{32} + 2A_{33} + A_{34}$  及  $M_{12} + M_{22} + M_{32} + M_{42}$ , 其中  $M_{ij}$  和  $A_{ij}$  分别是  $D$  中元素  $a_{ij}$  的余子式和代数余子式.

$$\text{解} \quad A_{31} + A_{32} + 2A_{33} + A_{34} = \begin{vmatrix} 3 & 1 & -1 & 2 \\ -5 & 2 & 3 & -4 \\ 1 & 1 & 2 & 1 \\ 1 & -5 & 3 & -3 \end{vmatrix} = \begin{vmatrix} 0 & -2 & -7 & -1 \\ 0 & 7 & 13 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & -6 & 1 & -4 \end{vmatrix} = \begin{vmatrix} -2 & -7 & -1 \\ 7 & 13 & 1 \\ -6 & 1 & -4 \end{vmatrix}$$

$$= -133,$$

$$M_{12} + M_{22} + M_{32} + M_{42} = -A_{12} + A_{22} - A_{32} + A_{42}$$

$$= \begin{vmatrix} 3 & -1 & -1 & 2 \\ -5 & 1 & 3 & -4 \\ 2 & -1 & 1 & -1 \\ 1 & 1 & 3 & -3 \end{vmatrix} = -14.$$

3. 解下列方程.

$$(1) \begin{vmatrix} a_1 - x & a_2 & \cdots & a_{n-1} & a_n \\ a_1 & a_2 - x & \cdots & a_{n-1} & a_n \\ \vdots & \vdots & & \vdots & \vdots \\ a_1 & a_2 & \cdots & a_{n-1} - x & a_n \\ a_1 & a_2 & \cdots & a_{n-1} & a_n - x \end{vmatrix} = 0;$$

$$(2) \begin{vmatrix} 1 & x & x^2 & \cdots & x^{n-1} \\ 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_{n-2} & a_{n-2}^2 & \cdots & a_{n-2}^{n-1} \\ 1 & a_{n-1} & a_{n-1}^2 & \cdots & a_{n-1}^{n-1} \end{vmatrix} = 0, \text{ 其中 } a_1, a_2, \cdots, a_{n-1} \text{ 互不相同.}$$

解 (1)

$$\begin{vmatrix} a_1 - x & a_2 & \cdots & a_{n-1} & a_n \\ a_1 & a_2 - x & \cdots & a_{n-1} & a_n \\ \vdots & \vdots & & \vdots & \vdots \\ a_1 & a_2 & \cdots & a_{n-1} - x & a_n \\ a_1 & a_2 & \cdots & a_{n-1} & a_n - x \end{vmatrix} \xrightarrow{c_1 + \sum_{j=2}^n c_j} \begin{vmatrix} a_1 + a_2 + \cdots + a_n - x & a_2 & \cdots & a_{n-1} & a_n \\ a_1 + a_2 + \cdots + a_n - x & a_2 - x & \cdots & a_{n-1} & a_n \\ \vdots & \vdots & & \vdots & \vdots \\ a_1 + a_2 + \cdots + a_n - x & a_2 & \cdots & a_{n-1} - x & a_n \\ a_1 + a_2 + \cdots + a_n - x & a_2 & \cdots & a_{n-1} & a_n - x \end{vmatrix}$$

$$= \begin{vmatrix} a_1 + a_2 + \cdots + a_n - x & a_2 & \cdots & a_{n-1} & a_n \\ 0 & -x & \cdots & 0 & 0 \\ \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & \cdots & -x & 0 \\ 0 & 0 & \cdots & 0 & -x \end{vmatrix} = (-x)^{n-1} (a_1 + a_2 + \cdots + a_n - x) = 0,$$

所以方程的根为 0 和  $a_1 + a_2 + \cdots + a_n$ .

$$(2) \begin{vmatrix} 1 & x & x^2 & \cdots & x^{n-1} \\ 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_{n-2} & a_{n-2}^2 & \cdots & a_{n-2}^{n-1} \\ 1 & a_{n-1} & a_{n-1}^2 & \cdots & a_{n-1}^{n-1} \end{vmatrix} = (a_1 - x)(a_2 - x) \cdots (a_{n-1} - x) \prod_{1 \leq j < i \leq n-1} (a_i - a_j) = 0$$

所以方程的根为  $a_1, a_2, \cdots, a_{n-1}$ .

4. 证明:

$$(1) \begin{vmatrix} x & -1 & 0 & 0 \\ 0 & x & -1 & 0 \\ 0 & 0 & x & -1 \\ a_4 & a_3 & a_2 & x + a_1 \end{vmatrix} = x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4;$$



$$(2) \begin{vmatrix} a_n & & & & b_n \\ & \ddots & & & \ddots \\ & & a_1 & b_1 & \\ & & c_1 & d_1 & \\ & \ddots & & & \ddots \\ c_n & & & & d_n \end{vmatrix} = \prod_{i=1}^n (a_i d_i - b_i c_i)$$

$$(3) \begin{vmatrix} \alpha + \beta & \alpha\beta & 0 & \cdots & 0 & 0 \\ 1 & \alpha + \beta & \alpha\beta & \cdots & 0 & 0 \\ 0 & 1 & \alpha + \beta & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \alpha + \beta & \alpha\beta \\ 0 & 0 & 0 & \cdots & 1 & \alpha + \beta \end{vmatrix} = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} \quad (\alpha \neq \beta);$$

证

$$(1) \begin{vmatrix} x & -1 & 0 & 0 \\ 0 & x & -1 & 0 \\ 0 & 0 & x & -1 \\ a_4 & a_3 & a_2 & x + a_1 \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4 & x^3 + a_1 x^2 + a_2 x + a_3 & x^2 + a_1 x + a_2 & x + a_1 \end{vmatrix}$$

$$= (-1)^{4+1} (x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4) \begin{vmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{vmatrix} = x^4 + a_1 x^3 + a_2 x^2 + a_3 x + a_4$$

(2) 将左边行列式记为  $D_{2n}$ ，把  $D_{2n}$  的第  $2n$  行依次与第  $2n-1$  行，第  $2n-2$  行， $\cdots$ ，第  $2$  行作相邻行的对换，得

$$D_{2n} = (-1)^{2n-2} \begin{vmatrix} a_n & 0 & \cdots & 0 & b_n \\ c_n & 0 & \cdots & 0 & d_n \\ 0 & a_{n-1} & & b_{n-1} & 0 \\ & \ddots & & & \\ \vdots & & a_1 & b_1 & \vdots \\ & & c_1 & d_1 & \\ & \ddots & & & \\ 0 & c_{n-1} & & d_{n-1} & 0 \end{vmatrix},$$

再把上面行列式的第  $2n$  列依次与第  $2n-1$  列，第  $2n-2$  列， $\cdots$ ，第  $2$  列作相邻列的对

换, 得

$$\begin{aligned}
D_{2n} &= (-1)^{2n-2} \cdot (-1)^{2n-2} \begin{vmatrix} a_n & b_n & 0 & \cdots & 0 \\ c_n & d_n & 0 & \cdots & 0 \\ 0 & 0 & a_{n-1} & & b_{n-1} \\ & & \ddots & \ddots & \\ \vdots & \vdots & & a_1 & b_1 \\ & & & c_1 & d_1 \\ & & \ddots & & \ddots \\ 0 & 0 & c_{n-1} & & d_{n-1} \end{vmatrix} \\
&= \begin{vmatrix} a_n & b_n \\ c_n & d_n \end{vmatrix} \begin{vmatrix} a_{n-1} & & & b_{n-1} \\ & \ddots & & \ddots \\ & & a_1 & b_1 \\ & & c_1 & d_1 \\ & \ddots & & \ddots \\ c_{n-1} & & & d_{n-1} \end{vmatrix} = (a_n d_n - b_n c_n) D_{2(n-1)} \\
&= (a_n d_n - b_n c_n)(a_{n-1} d_{n-1} - b_{n-1} c_{n-1}) D_{2(n-2)} \\
&\quad \dots\dots\dots \\
&= (a_n d_n - b_n c_n)(a_{n-1} d_{n-1} - b_{n-1} c_{n-1}) \cdots (a_2 d_2 - b_2 c_2) D_2
\end{aligned}$$

又  $D_2 = \begin{vmatrix} a_1 & b_1 \\ c_1 & d_1 \end{vmatrix} = a_1 d_1 - b_1 c_1$ , 故

$$\begin{aligned}
\text{左边} &= (a_n d_n - b_n c_n)(a_{n-1} d_{n-1} - b_{n-1} c_{n-1}) \cdots (a_2 d_2 - b_2 c_2)(a_1 d_1 - b_1 c_1) = \prod_{i=1}^n (a_i d_i - b_i c_i) \\
&= \text{右边}.
\end{aligned}$$

(3) 将左边行列式记为  $D_n$ , 按第 1 列展开得

$$D_n = (\alpha + \beta) D_{n-1} - \alpha \beta D_{n-2}$$

即

$$D_n - \alpha D_{n-1} = \beta (D_{n-1} - \alpha D_{n-2}),$$

此式对一切  $n$  都成立, 故递推得

$$\begin{aligned}
D_n - \alpha D_{n-1} &= \beta^2 (D_{n-2} - \alpha D_{n-3}) = \beta^3 (D_{n-3} - \alpha D_{n-4}) \\
&= \cdots = \beta^{n-2} (D_2 - \alpha D_1) = \beta^{n-2} ((\alpha + \beta)^2 - \alpha \beta - \alpha(\alpha + \beta)) = \beta^n,
\end{aligned}$$

在原式中,  $\alpha, \beta$  的地位是一样的, 交换  $\alpha, \beta$  的位置, 同理可得

$$D_n - \beta D_{n-1} = \alpha^n .$$

解方程

$$\begin{cases} D_n - \alpha D_{n-1} = \beta^n \\ D_n - \beta D_{n-1} = \alpha^n \end{cases} ,$$

得

$$D_n = \frac{\alpha^{n+1} - \beta^{n+1}}{\alpha - \beta} = \text{右边} .$$